

Tribhuvan University
Institute of Science and Technology

2079 (Set A)

Master Level / First Year / Second Semester / Science

Computer Science and Information Technology (CSc 558)

(Compiler Optimization)

Full Marks : 45

Pass Marks : 22.5

Time : 2 Hrs

Group A

Attempt any **TWO** questions

(2 × 10 = 20)

1. Why do we need to optimize the code in compiler? Differentiate between loop peeling and section based splitting with example. [2 + 8]
2. Define forward, backward and exit branch. Remove the forward branch from the following instruction and vectorize it. [3 + 7]

DO I = 1, N

IF (A(I) > 10) GOTO 60

20 A(I) = A(I) + 10

IF (B(I) > 10) GOTO 80

40 B(I) = B(I) + 10

60 A(I) = B(I) + A(I)

80 B(I) = A(I) - 5

END DO

3. Show the dependence graph and remove the dependence using array renaming for the following instruction. [10]

DO I=1, N

S1: A(I) = A(I-1) + X

S2: Y(I) = A(I) + Z

S3: A(I) = B(I)

END DO

$A[1] = A[0] + X$
 $Y[1] = A[1] + Z$
 $A[1] = B[1]$

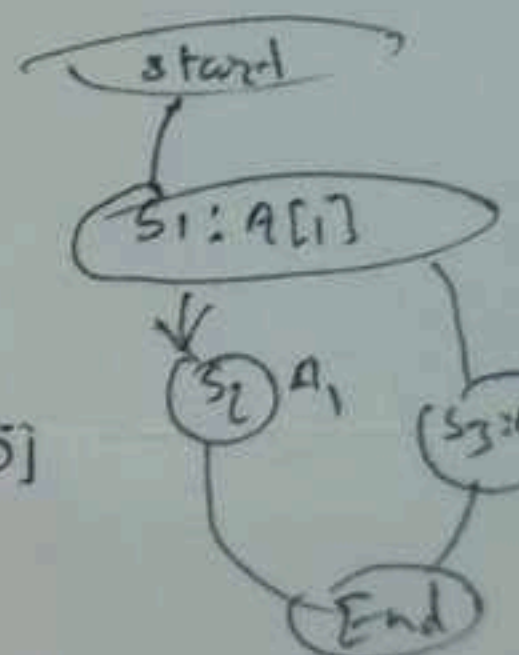
$A[2] = A[1] + X$
 $Y[2] = A[2] + Z$
 $A[2] = B[2]$

Group B

Attempt **ALL** the questions

(5 × 5 = 25)

4. Describe any two types of data dependence with example. [5]
5. How do you compute distance and direction vector? Illustrate with your own example. [5]
6. Discuss about constant propagation. [5]
7. What is the purpose of loop skewing? Explain. [5]
8. Define induction variable? Give a scenario to express the control dependence in loop. [1+4]



Master Level / First Year/ IInd Semester/ Science
Computer Science and Information Technology (CSc. 564)
(Data Warehousing and Data Mining)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group A

Attempt any TWO questions.

(2×10=20)

1. Differentiate between supervised and unsupervised algorithm. Explain KDD process with suitable figure. (2+8)
2. Define data warehouse. Suppose that a data warehouse consists of the four dimensions {location, project leader, project, client}. Similarly, it has two measures average_cost and rating, where rating is the value that a project leader provides to a project based on its status. (2+4+4)

Draw a schema diagram for the above data warehouse using

- a) Star schema
- b) Snowflake schema

3. Use ID3 algorithm to construct the decision tree on the following data set. (10)

ID	Condition	Price	Brand	Buy
1	good	high	A	yes
2	bad	medium	B	no
3	bad	low	A	no
4	good	high	C	yes
5	good	low	A	no

Group B

Attempt ALL questions.

(5×5=25)

4. Explain the architecture of data mining system. (5)
5. What are the demerits of Apriori algorithm? How it can be solved? (1+4)
6. Explain any one PAM method for clustering. (5)
7. Explain multilevel association mining. What is zero-probability error and how it can be handled? (2+1+2)
8. What do you mean by spatial database? Explain the need for mining spatial database. (1+4)

Master Level / First Year/ IInd Semester/ Science
Computer Science and Information Technology (CSc. 559)
(Web Systems and Algorithm)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours.

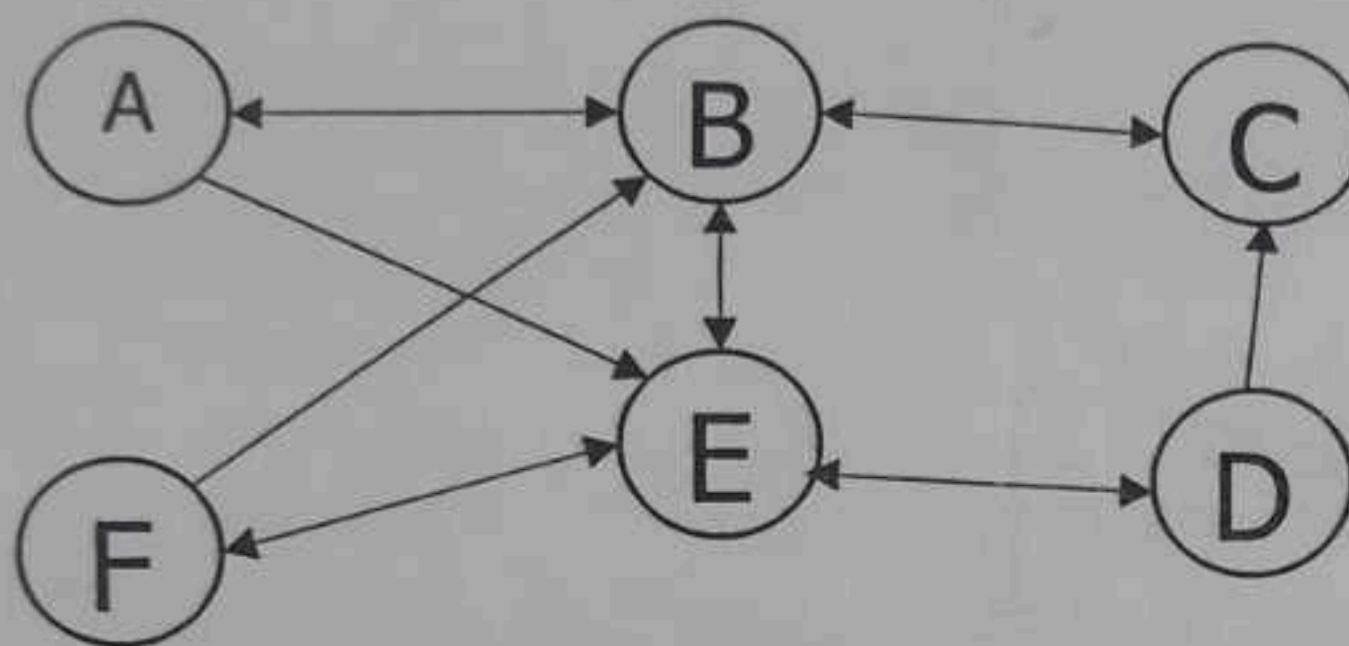
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group A

Attempt any two questions.

(2×10=20)

1. Compare Page Rank algorithm with HITS algorithm. Use Page-Rank algorithm in the graph below up to iteration 2. Use damping factor 0.9.



2. What is user based collaborative filtering? Suppose, the ratings of five users are indicated for six items. Each rating is drawn from the range {1 ... 5}. Find the predictions of user "User2" for items "Item3" and "Item5" on the basis of the ratings in the table below using user-based collaborative filtering recommender system. Use cosine similarity to find similarity between users and consider top-2 closest users.

User/Item	Item1	Item2	Item3	Item4	Item5	Item6
User1	3	4	5	4	4	4
User2	4	5	?	4	?	4
User3	?	3	3	1	1	3
User4	2	2	1	3	3	4
User5	2	?	4	2	3	3

3. How can we use classification algorithms for intelligent web applications? Explain ROCK clustering algorithm in detail.

Group B

(5×5=25)

Attempt ALL questions.

4. Explain basic elements of intelligent applications in detail.
5. Why link analysis is important in searching? What is association rule mining?
6. How can you evaluate performance of a classifier? What is bagging?
7. Define semantic web. Explain RDF with example.
8. Write short notes on:
 - a. Content-based recommendation
 - b. Semantic web technologies

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Master Level / First Year/ IInd Semester/ Science
Computer Science and Information Technology (CSc. 561)
(Machine Learning)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group A

Attempt any two questions.

(2×10=20)

1. Consider the following dataset. Find Information gain of each attribute and construct decision tree. then, predict class label for the tuple {Cloudy, Low, Normal}.

Weather	Temperature	Wind Level	Go Out (Class)
Sunny	High	Low	No
Sunny	Normal	Normal	Yes
Cloudy	High	Normal	No
Cloudy	Normal	High	Yes
Sunny	Normal	High	No
Rainy	High	Normal	No
Rainy	Low	High	No

2. We have to divide the data points {2, 4, 6, 10, 14, 16, 18} into two clusters using GMM. Estimate parameters after first iteration of execution.
3. Derive SVM optimization Problem.

Group B

Attempt all questions.

(5×5=25)

4. What is machine learning? How is it works? Explain.
5. Derive weight update rule for logistic regression.
6. Consider following confusion matrix and the calculate accuracy, weighted average accuracy and macro and micro precision, recall, and F1-score.

		Actual Class		
		Apple	Orange	Mango
Predicted Class	Apple	7	8	9
	Orange	1	2	3
	mango	3	2	1